

Project Title: Project Management System (Salesforce Flow Automation)

Problem Statement

In a dynamic organization, managing multiple projects simultaneously can become complex and time-consuming. Manually creating project templates, assigning tasks, tracking progress, and ensuring standardization across teams often leads to **inconsistency, delays, and data duplication**.

The challenge is to build a **scalable, automated Project Management System** in **Salesforce** that simplifies task creation, project tracking, and standard workflow execution — all within a few clicks.

Proposed Solution

This **Project Management System** automates the creation of projects and their associated tasks using **Salesforce Flows, Custom Objects, and Record Templates**.

By leveraging the **Project Template** concept, this solution allows users to:

- Create reusable templates for projects and task structures.
- Automatically generate new projects with predefined task sets.
- Streamline assignments using flows and collections.
- Maintain control and access via custom permission sets.

This reduces repetitive manual work, ensures consistent project execution, and enhances productivity within Salesforce.

System Workflow

1. **User selects a Project Template.**
2. **Flow triggers** to create a new Project record.
3. **Associated Task Templates** are automatically cloned under the new Project.
4. The flow handles **task assignments, looping, and debug validation** for accuracy.
5. **Permission Sets** ensure only authorized users can access, modify, or trigger templates.

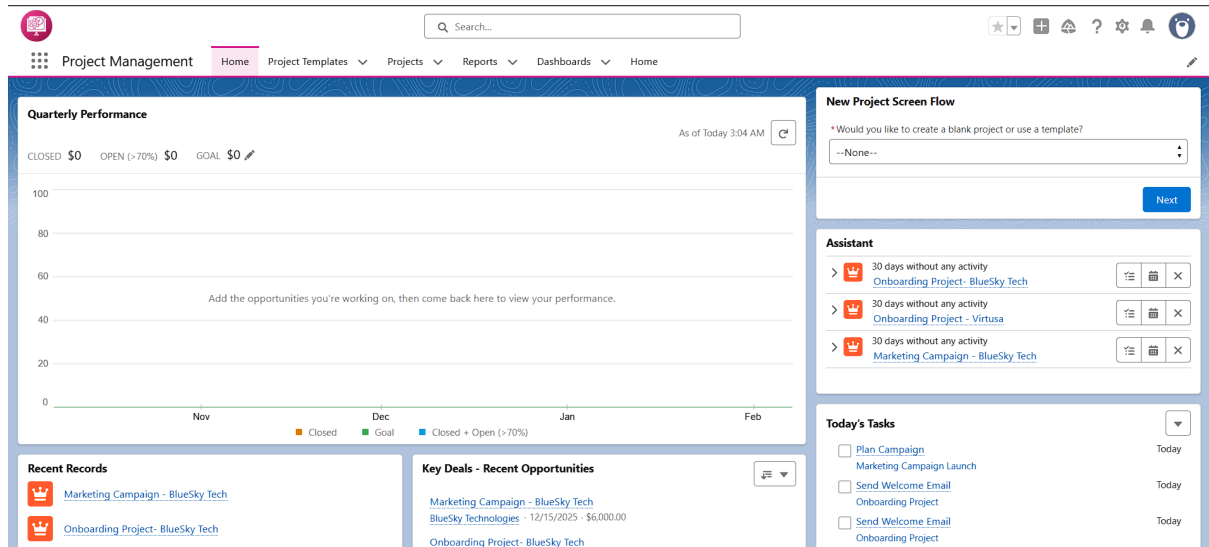
This workflow provides an automated project setup process — minimizing manual configuration.

Objects Used

1. **Project__c** – Represents the main project record, containing project name, owner, and key details.
2. **Project_Template__c** – Stores reusable project structures. Each template defines default project attributes and associated task templates.
3. **Task_Template__c** – Defines standard tasks under each project template, with task details such as subject, status, and priority.
4. **Project_Task__c** – Represents actual tasks created for a specific project based on task templates.
5. **Flow Variables & Collections** – Used for automation logic (loops, assignments, record creation).
6. **Permission Set** – Controls access rights for users managing templates and flow executions.

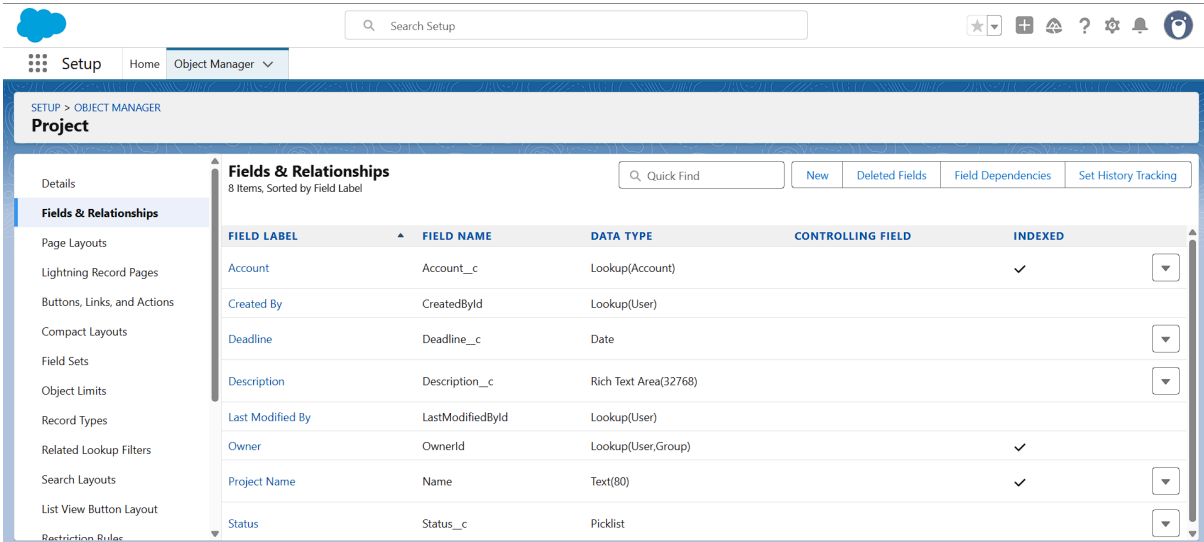
Detailed Workflow with Screenshots

1. Project App Home Screen



The main Project Management App interface where users can access templates, projects, and tasks.

2. Project Object Fields



The screenshot displays the Project Object Fields configuration screen. The top navigation bar includes a search bar and a menu with options: Setup, Home, and Object Manager. The main content area is divided into several sections:

- Fields & Relationships:** A table showing the configuration of fields and relationships for the Project object. The table has 5 columns: FIELD LABEL, FIELD NAME, DATA TYPE, CONTROLLING FIELD, and INDEXED. The table contains 8 rows of data.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Account	Account_c	Lookup(Account)		✓
Created By	CreatedById	Lookup(User)		
Deadline	Deadline_c	Date		
Description	Description_c	Rich Text Area(32768)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Project Name	Name	Text(80)		✓
Status	Status_c	Picklist		

Custom fields created for the Project object to store relevant project information such as owner, start date, and status.

3. Project Template Object Fields

The screenshot shows the Salesforce Setup interface for the Project Template object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'Project Template' and shows the 'Fields & Relationships' section with 6 items, sorted by Field Label. The table lists the following fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Active	Active__c	Checkbox		
Created By	CreatedById	Lookup(User)		
Description	Description__c	Rich Text Area(32768)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Project Template Name	Name	Text(80)		✓

Defines reusable templates for projects. Each template can predefine project properties for faster project creation.

4. Project Task Template Object Fields

The screenshot shows the Salesforce Setup interface for the Project Task Template object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Restriction Rules, Scoping Rules, and Object Access. The main content area is titled 'Project Task Template' and shows the 'Fields & Relationships' section with 6 items, sorted by Field Label. The table lists the following fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Description	Description__c	Long Text Area(32768)		
Due Date Offset	Due_Date_Offset__c	Number(18, 0)		
Last Modified By	LastModifiedById	Lookup(User)		
Project	Project__c	Master-Detail(Project Template)		✓
Task Name	Name	Text(80)		✓

Specifies the fields for standard task templates linked to a project template, including task subject and priority.

5. Project Template (Version 1) with Tasks

Project Management

Home

Project Templates

Projects

Reports

Dashboards

Home

Q Search...

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Project Template

Onboarding Project

New Contact

Edit

New Opportunity

Details

Project Template Name

Onboarding Project

Owner

Sarathi Thokal

Description

Make sure all new clients feel comfortable with our process and are familiar with the resources available to them.

Active

☒

Created By

Sarathi Thokal · 11/7/2025, 10:23 AM

Last Modified By

Sarathi Thokal · 11/7/2025, 2:03 PM

Tasks (3)

3 items • Sorted by Due Date Offset • Updated a few seconds ago

☐

Task Name

Due Date Offset

Description

1

☐ Send Welcome Email

0

Refer to the template: "New Client Welcome"

2

☐ Schedule First Meeting

7

3

☐ Complete Post Sale Interview

14

Refer to the survey: "Post Sale Interview"

Shows how tasks are associated with a project template for standardized structure.

6. Project Template (Version 2) with Tasks

Project Management

Home

Project Templates

Projects

Reports

Dashboards

Home

Q Search...

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Refer to the template: "New Client Welcome"

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7

3

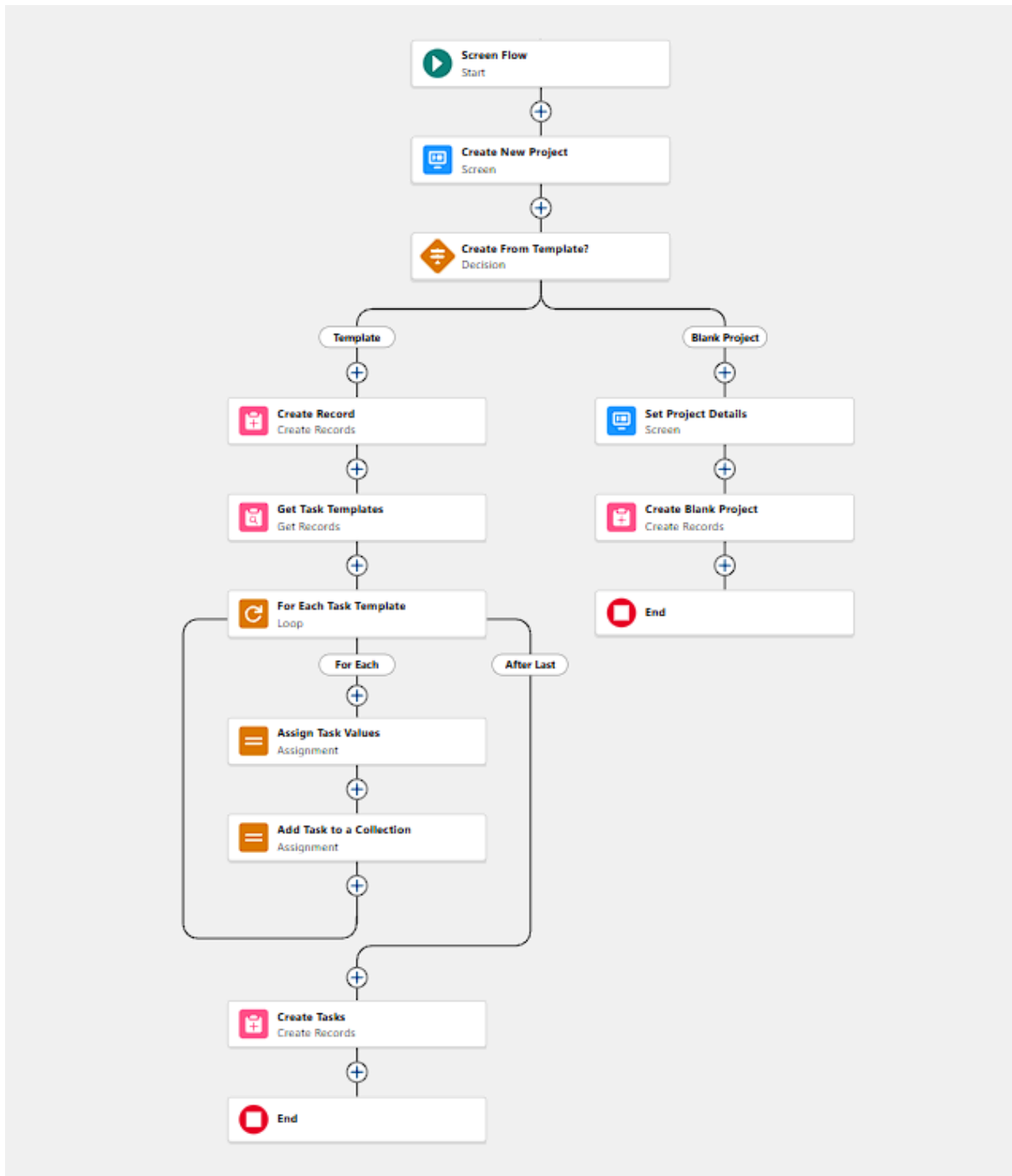
☐ Complete Post Sale Interview

14

Refer to the survey: "Post Sale Interview"

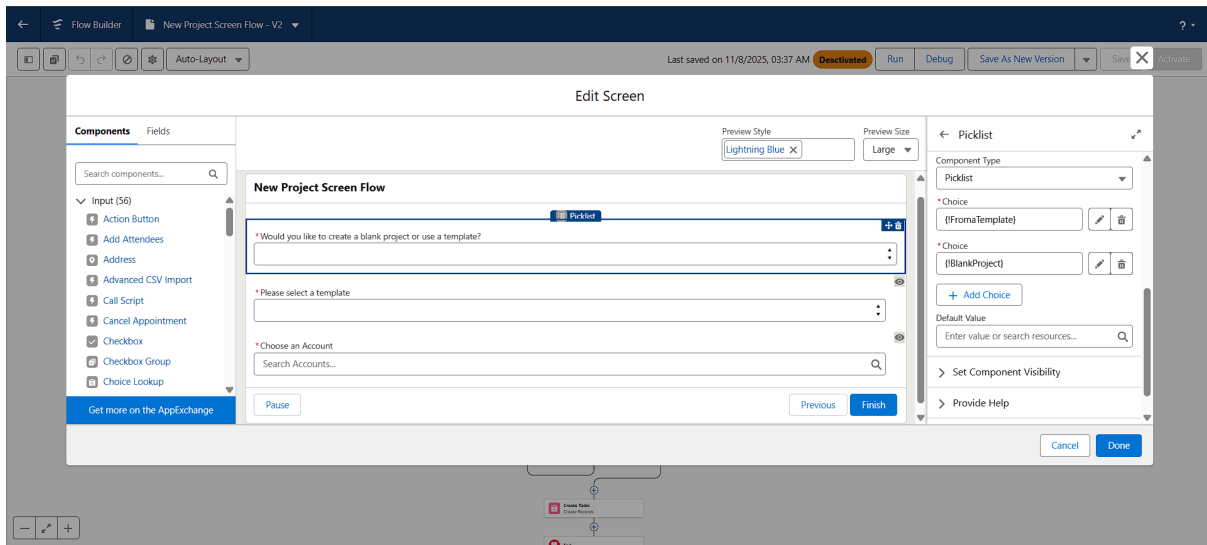
A second version of the project template illustrating different task arrangements or stages.

7. Flow Overview



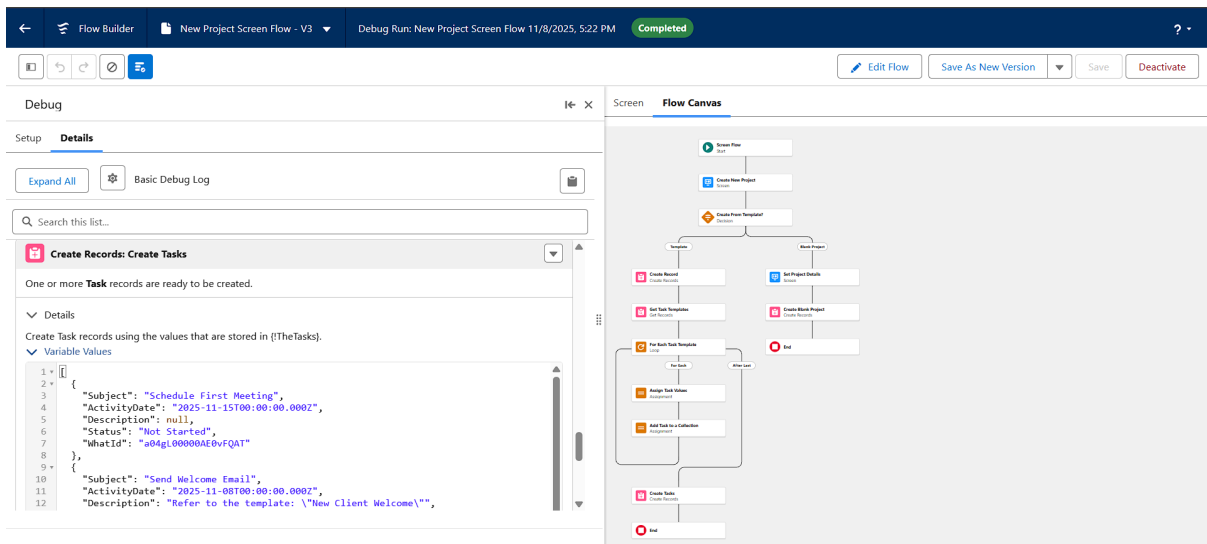
Visual representation of the Salesforce Flow designed to automate project and task creation.

8. Project Template with Tasks (Final View)



Final version of the Project Template displaying its relationship with task templates.

9. Flow Debug Success – Template 1



Successful execution of the flow with Template 1, validating that all records were created correctly.

10. Flow Debug Success – Template 2

Debug

Setup Details

Expand All Basic Debug Log

Search this list...

Result

All records are ready to be created when the next screen, pause, or local action is executed or when the interview finishes.

How the Interview Finished

Details

The flow interview ran for 14.11 seconds and finished on November 8, 2025 at 5:22 PM.

Screen Flow Canvas

Flow diagram showing steps: Screen Flow, Create New Project, Create Project Template, Set Project Details, Get Task Templates, Create Blank Project, For Each Task Template, Assign Task Values, Add Task to a Collection, Create Tasks, End.

Debug output for Template 2 showing correct looping and record creation.

11. Flow Loop Assignment

Flow Canvas

Assignment

Label: Assign Task Values

API Name: Assign_Task_Values

Description:

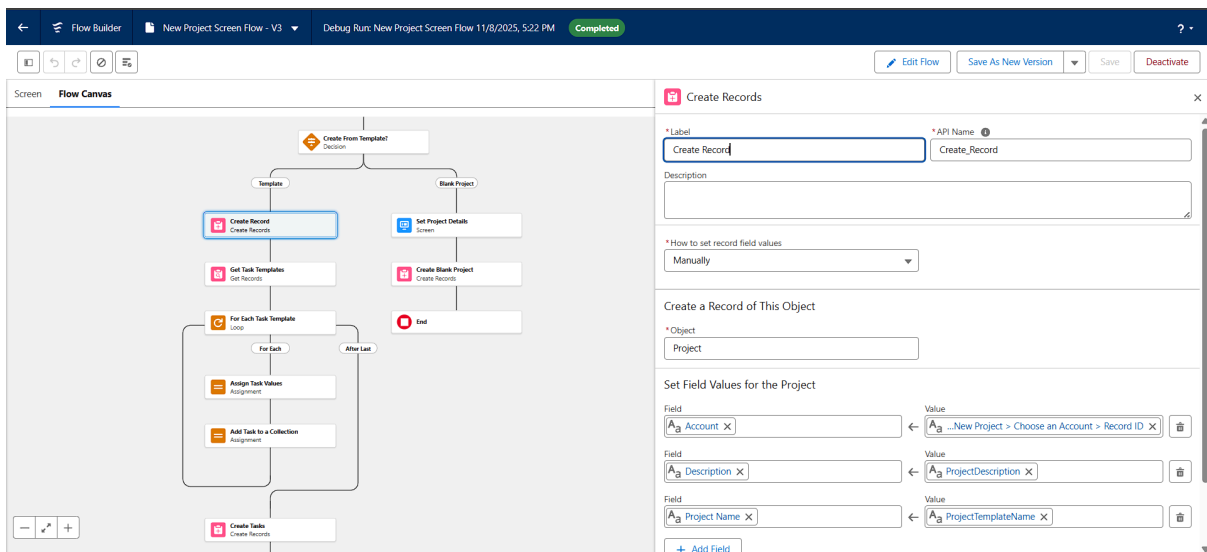
Set Variable Values

Each variable is modified by the operator and value combination.

Variable	Operator	Value
Task > Subject	Equals	...ch Task Template > Task Name
Task > Due Date Only	Equals	DueDateFormula
Task > Description	Equals	...ch Task Template > Description
Task > Status	Equals	Not Started
Task > Related To ID	Equals	...rom Create Record > Record ID

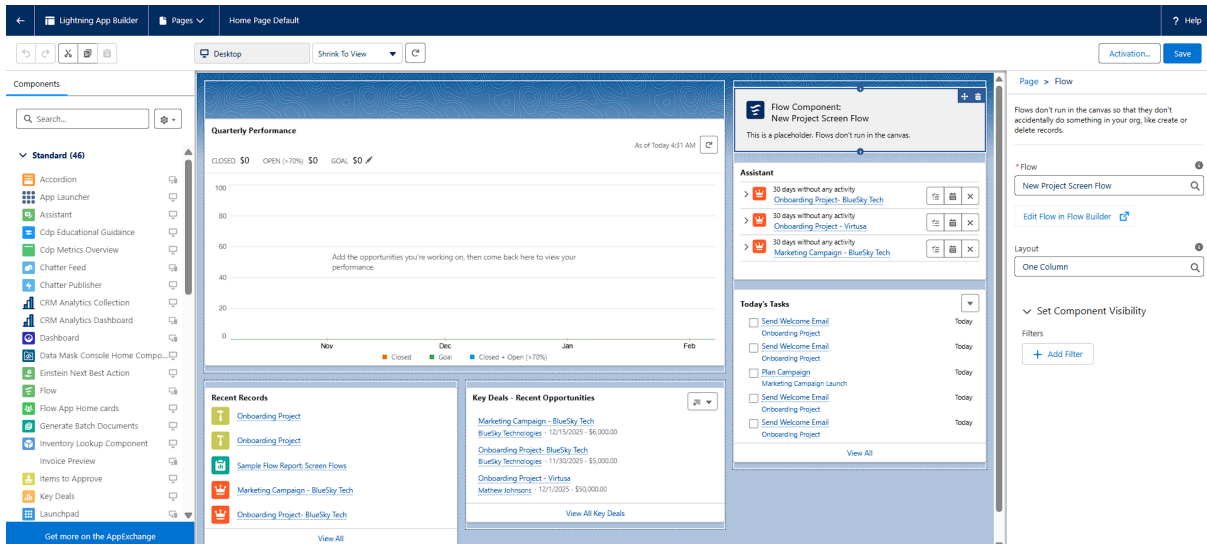
Part of the automation where the flow loops through task templates to create individual project tasks.

12. Flow Create Tasks Collection



Flow element that stores created task records in a collection variable before committing to the database.

13. Flow Home Component



Main component of the flow that initiates the process for creating a new project from templates.

14. Permission Set (View 1)

SETUP Profiles									
Custom Object Permissions									
	Basic Access				Data Administration				
	Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields		
Leave Requests	✓	✓	✓	✓	✓	✓	✓		
Mentors	✓	✓	✓	✓	✓	✓	✓		
Projects	✓	✓	✓	✓	✓	✓	✓		
Project Task Templates	✓	✓	✓	✓	✓	✓	✓		
Project Templates	✓	✓	✓	✓	✓	✓	✓		
Students	✓	✓	✓	✓	✓	✓	✓		
Trainings	✓	✓	✓	✓	✓	✓	✓		

Shows permission assignments to control which users can access and execute flows.

15. Permission Set (View 2)

SETUP Profiles									
Standard Object Permissions									
	Basic Access				Data Administration				
	Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields		
Accounts	✓	✓	✓	✓	✓	✓	✓		

Another view of the permission set configuration for more granular access control.

Key Features

- Automated Project & Task Creation
- Intelligent Salesforce Flow Logic
- Template-Based Reusability
- Permission-Controlled Access
- Consistent and Scalable Project Management



Future Enhancements

- Add **task dependencies** and **progress tracking**.
 - Integrate **email notifications** for task assignments.
 - Implement a **dashboard** for project overview and analytics.
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Tech Stack

- **Platform:** Salesforce
 - **Automation:** Flows
 - **Objects:** Custom Salesforce Objects
 - **Access Control:** Permission Sets
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Conclusion

This Salesforce-based Project Management System demonstrates how automation and reusable templates can **streamline workflows**, **reduce errors**, and **enhance consistency** in project execution.

By combining custom objects, flows, and permission sets, it delivers a flexible, scalable solution adaptable to any organization's project needs.